

SQLintersection

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3:00pm - 4:00pm

SQL SERVER 2016/2017 ALWAYSON AVAILABILITY GROUPS: WINDOWS AND LINUX DEPLOYMENTS

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SQL
intersection



Overview

■ Introduction

- Planning and Deploying AlwaysOn Availability Groups
- Readable Secondary Replicas and Database Maintenance
- AlwaysOn Integration with Windows and Linux
- Management and Monitoring of Availability Groups
- SQL Server 2016 / 2017 Improvements and Features

■ Technologies Covered

- AlwaysOn Availability Groups
- Azure Portal
- Linux / Pacemaker

Causes of Downtime and Data Loss

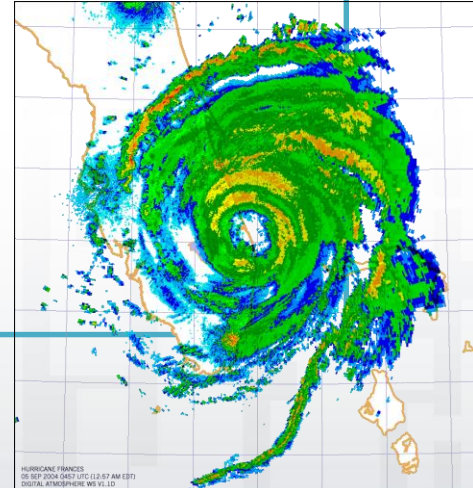
Planned Downtime

- Maintenance
- Upgrade
- Updates
- No data loss



Unplanned Downtime

- Datacenter failure
- Server failure
- I/O subsystem failure
- Human error
- Possible data loss



High Availability and Disaster Recovery Concepts

Understanding Disaster Recovery



Two Principle Requirements for HA and DR

- **Recovery Time Objective (RTO)** - The maximum allowable downtime when a failure occurs
- **Recovery Point Objective (RPO)** - The maximum allowable data-loss when a failure occurs



High Availability and Disaster Recovery Concepts

Introduction to AlwaysOn Technologies



SQL Server AlwaysOn Solutions

Failover Cluster Instances (FCI)

- Enhancements
 - Cloud Witness
 - Site Awareness
 - Fault Domains
- SQL Server **Instance Failover**
- Shared Storage
- Passive Secondary Nodes
- Failover takes 30s to couple of minutes (instance restart)

Availability Group (AG)

- Introduced in SQL 2012
- **Multi-database failover**
- Non Shared Storage
- Active Secondary Replicas
- Failover takes less than 30s (secondary replicas are online)
- New Features introduced in subsequent SQL Server versions

SQL Server AlwaysOn Solutions

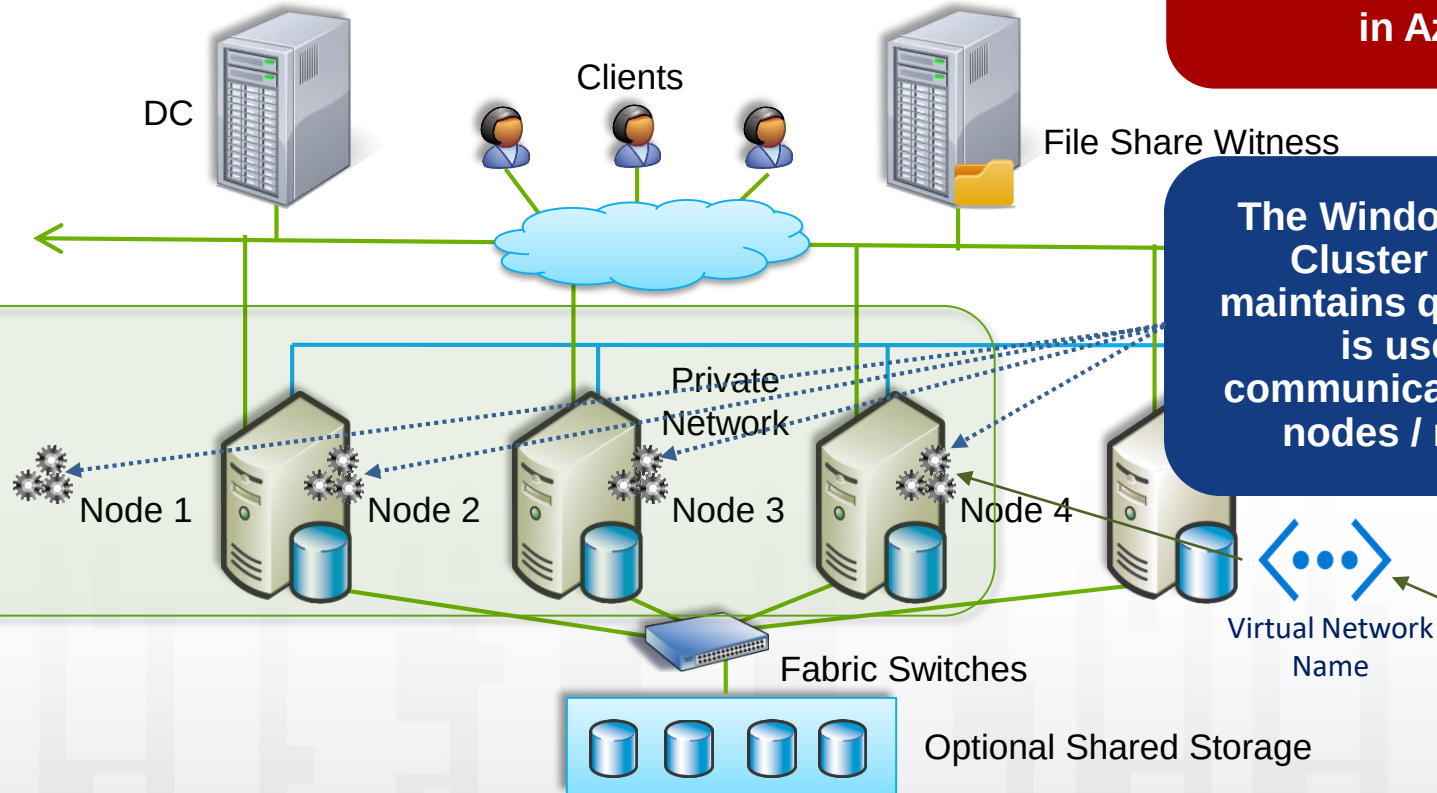
- **Cloud Witness** - Allows you to leverage Azure Blob Storage to act as a witness for your cluster in place of a Disk Witness or File Share Witness

This enables these scenarios:

- a). Failover Cluster in Azure, b). Branch Office Clusters, c). Multi-site Clusters
- **Site Awareness** - Allows you to control the failover to always prefer the local node. This helps when failing over to another site / DR site could cause an unexpected issue of having to failover again back to a preferred node
- **Fault Domains** - Lets you define Node, Chassis, and Rack locations in addition to Site
 - Fault Domains have three benefits:
 - Storage Affinity in a Stretch Cluster
 - Ensures Cluster workload and storage are in the same location
 - Increases Storage Spaces resiliency
 - Enhanced Health Service Alerts include meta data about the location of the associated resources raising the alarm

Overview of a Windows Server Failover Cluster

Windows Server 2016 allows you to create a cluster without a Domain and a Witness in Azure



Reviewing Availability Groups

AlwaysOn Availability Groups and Replicas



AlwaysOn Availability Groups

AlwaysOn **Availability Groups** is a new feature that enhances and combines database mirroring, clustering, and log shipping capabilities

Flexible

- Multi-database failover
- Multiple secondaries
 - Total of 4 secondaries (8 with 2014+)
 - 3 synchronous secondaries*
 - 2 automatic failover pair**
- Synchronous and asynchronous
- Built in compression and encryption
- Auto-page repair
- Automatic & manual failover
- Flexible failover policy
- Create an availability group w/o a cluster for read-scale workloads ¥

Integrated

- Application failover using virtual name
- Configuration Wizard
- AlwaysOn Management Dashboard
- System Center Integration
- Rich diagnostic infrastructure
- File-stream replication
- Replication publisher failover
- Read-only routing **
- Guarantee commits on synchronous secondary replicas ¥
- Support for SQL Server on Linux ¥

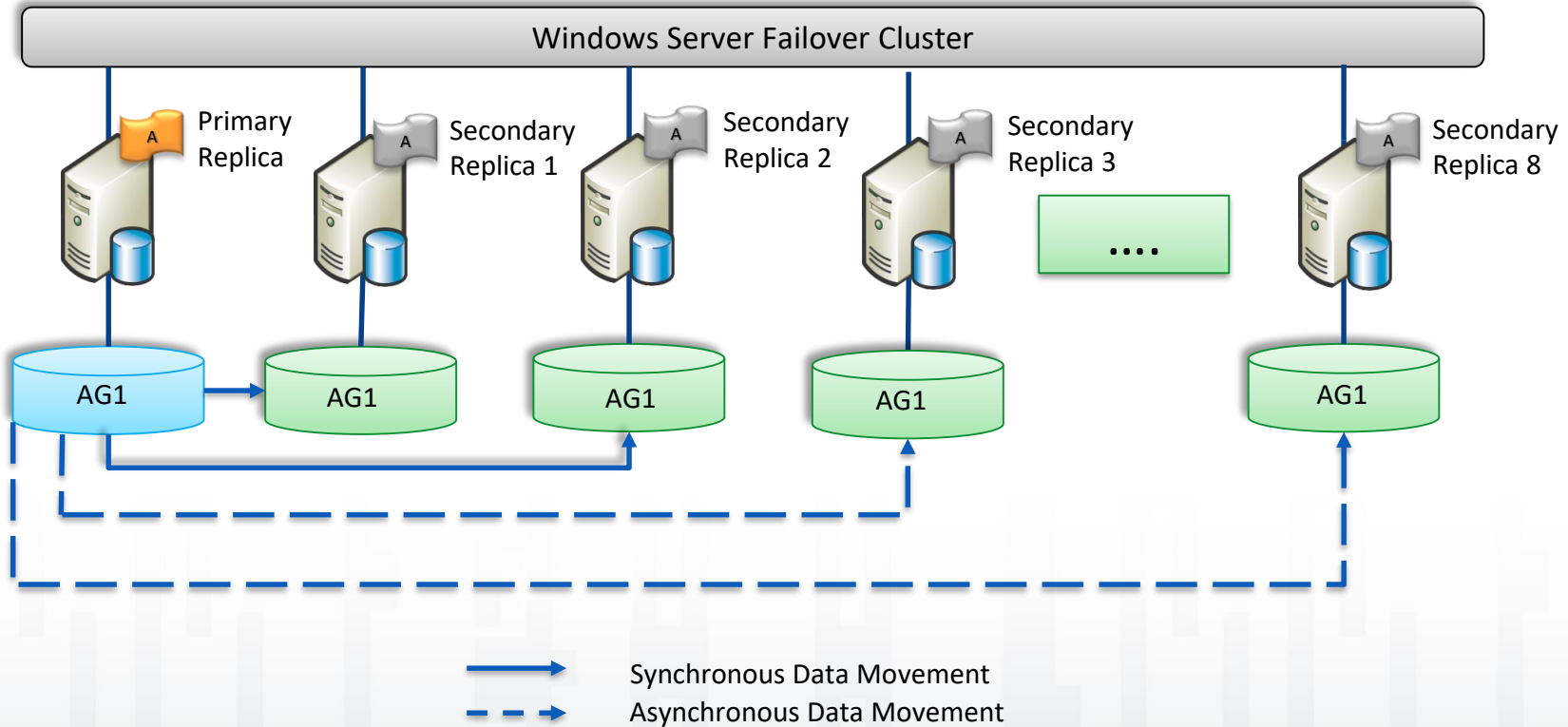
Efficient

- Active Secondary
 - Readable Secondary
 - Backup from Secondary
 - Improves primary server performance by offloading work to secondary
- Monitoring & Troubleshooting enhanced
- Automation using PowerShell

* New to SQL Server 2014 ¥ New to SQL Server 2017

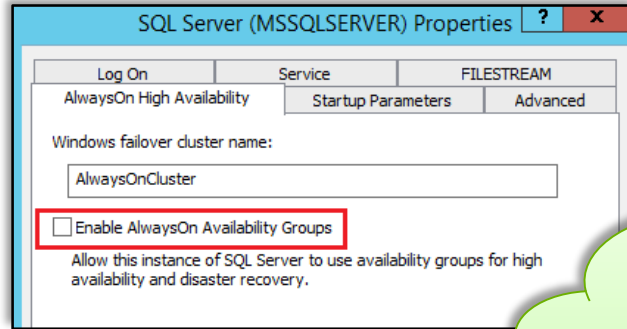
** New to SQL Server 2016

AlwaysOn Availability Groups and Replicas



Enabling AlwaysOn Availability Groups

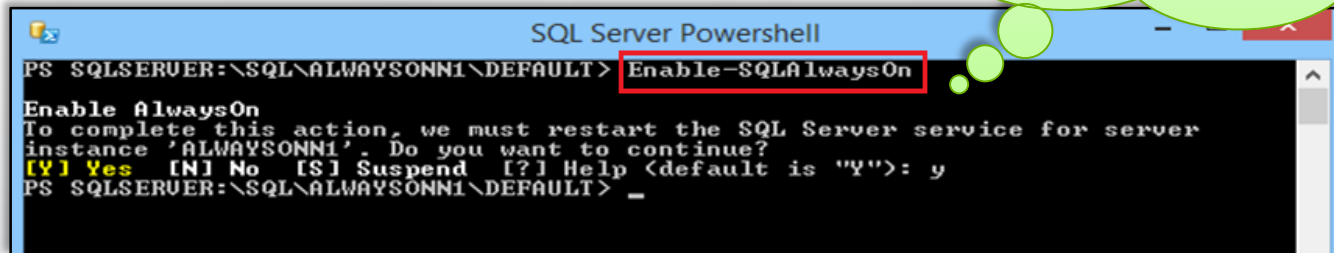
Use **SQL Server Configuration Manager**



You must first install failover clustering and put the replicas in the same cluster

Requires 'Windows Management Instrumentation (ASync-In)' Firewall rule to be enabled

or **Enable-SQLAlwaysOn** Windows PowerShell cmd



Availability Groups in SQL Server 2014

- Increased Number of Secondaries
- **SQL 2014 increased to 8 replicas**
- Additional Diagnostic Capabilities (fn_hadr* function, cluster DMVs)
- Increased Readable Secondaries Availability
 - Reduced the events which caused the readable secondaries to be unavailable
- **Add Azure Replica Wizard**
 - Simple end-to-end solution for deploying replicas to Azure VMs
 - First step to improved deployment experiences
 - **Available via T-SQL, but currently removed from SSMS**

AlwaysOn Availability Group Availability Modes

Asynchronous-Commit

- Disaster-recovery solution
- Works well when replicas are distributed over considerable distances
- Useful when performance is more important than synchronized data protection

Synchronous-Commit

- High availability over performance at the cost of increased transaction latency

Different ways to Create Availability Groups

SQL Server Management Studio

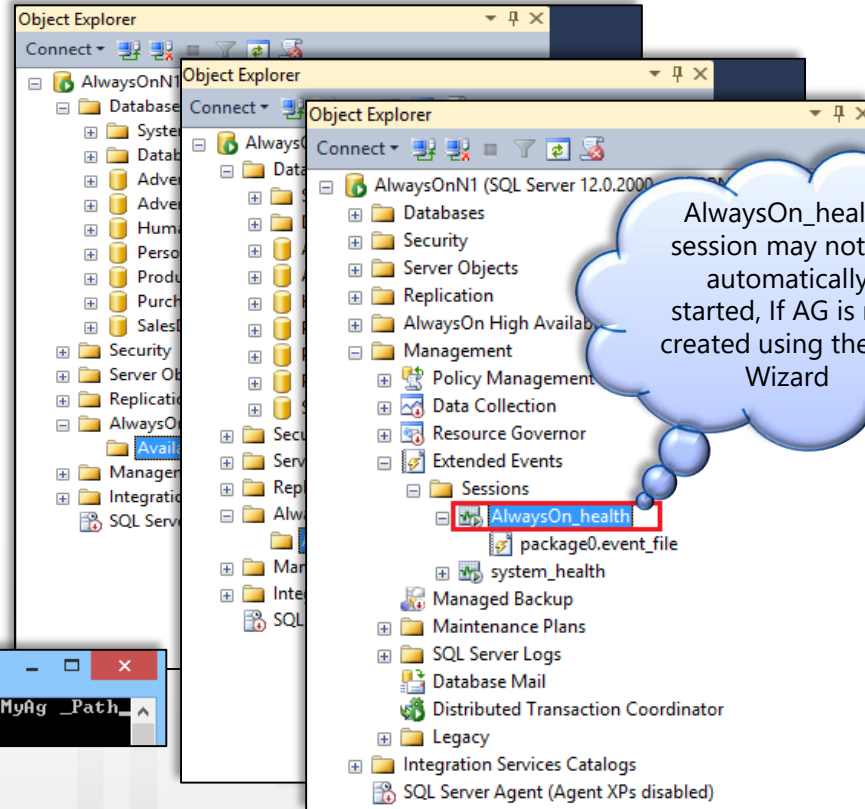
- **Availability Group Wizard**
- **Availability Group Dialog Box**

T-SQL

```
SQLQuery2.sql - AI...NET\cladmin (53))*  
CREATE AVAILABILITY GROUP MyAg
```

PowerShell

```
SQL Server Powershell  
PS SQLSERVER:\SQL\ALWAYSONN1\DEFAULT> New-SqlAvailabilityGroup -Name MyAg -Path_
```



Demo

AlwaysOn Availability Groups

- Creating AlwaysOn Availability Groups
- Failing Over an Availability Group
- Examining the AlwaysOn AG Dashboard

Flexible Failover Policy and Health Monitoring

- **sp_server_diagnostics** stored procedure called to perform a health check of the primary replica
- Results at an interval that equals $1/3$ of the health-check timeout threshold
- Default health-check timeout threshold is 30 seconds
 - sp_server_diagnostics returns results at 10 second interval

sp_server_diagnostics

- Captures diagnostic data and health information about SQL Server to detect potential failures
- Runs in repeat mode and sends results periodically
- Sample Result:

The diagram illustrates the output of the `sp_server_diagnostics` stored procedure. It features a table with five rows of data. Callouts provide context for specific parts of the table:

- A callout points to the first three rows (system, resource, query_processing), stating: "System, resource and query_processing leveraged for failure detection".
- A callout points to the last two rows (io_subsystem, events), stating: "io_subsystem and events are leveraged for diagnostic purposes only".
- A callout points to the entire table, stating: "5 Components".
- A callout points to the 'state' and 'state_desc' columns, stating: "Health State".

create_time	component_ty...	component_name	state	state_desc	data
2014-10-16 ...	instance	system	1	clean	<sys...
2014-10-16 ...	instance	resource	1	clean	<res...
2014-10-16 ...	instance	query_processing	1	clean	<qu...
2014-10-16 ...	instance	io_subsystem	1	clean	<ioS...
2014-10-16 ...	instance	events	0	unknown	<eve...

Failure Condition Levels

5 – Failover if any qualified failure conditions	Query Processing errors
4 – Failover if moderate SQL Server errors	Resource errors
3 – Failover if critical SQL Server errors (Default)	System errors
2 – Failover if server is unresponsive	No response from sp_server_diagnostics
1 – Failover if server is down or lease timeout	Service is down

Note: Damaged databases and suspect databases are not detected by any failure-condition level. Therefore, a database that is damaged or suspect (whether due to a hardware failure, data corruption, or other issue) never triggers an automatic failover.

Demo

AlwaysOn Availability Groups

- `sp_server_diagnostics` check
- Flexible Failover Policy

AlwaysOn Availability Group Monitoring

Dashboards, System Views,
Wait Statistics and Performance Monitor



Dashboard

Open from
primary

Group
health

Replica
health

Database
health

AGCorp:AlwaysOnN1

AGCorp: hosted by ALWAYSONN1 (Replica role: Primary)

Availability group state: Healthy
Primary instance: ALWAYSINN1
Failover mode: Automatic
Cluster state: AlwaysOnCluster (Normal Quorum)

Availability replica:

Name	Role	Failover Mode	Synchronization State	Availability M...	Issues
ALWAYSONN1	Primary	Automatic	Synchronized	Synchronous c...	
ALWAYSONN2	Secon...	Automatic	Synchronized	Synchronous c...	
ALWAYSONN3	Secon...	Manual	Synchronizing	Asynchronous ...	

Group by

Name	Replica	Synchronization State	Failover Read...	Issues
ALWAYSONN1				
AdventureWorks2014	ALWAYSONN1	Synchronized	No Data Loss	
AdventureWorksDW2...	ALWAYSONN1	Synchronized	No Data Loss	
ALWAYSONN2				
AdventureWorks2014	ALWAYSONN2	Synchronized	No Data Loss	
AdventureWorksDW2...	ALWAYSONN2	Synchronized	No Data Loss	
ALWAYSONN3				
AdventureWorks2014	ALWAYSONN3	Synchronizing	Data Loss	
AdventureWorksDW2...	ALWAYSONN3	Synchronizing	Data Loss	

Name

Replica

Synchronization State

Failover Readiness

Issues

Suspended

Suspend Reason

Estimated Recovery Time (seconds)

Estimated Data Loss (time)

Synchronization Performance (seconds)

Log Send Queue Size (KB)

Log Send Rate (KB/sec)

Redo Queue Size (KB)

Redo Rate (KB/sec)

FileStream Send Rate (KB/sec)

End of Log LSN

Recovery LSN

Truncation LSN

Last Commit LSN

Last Commit Time

Last Sent LSN

Last Sent Time

Last Received LSN

Last Received Time

Last Hardened LSN

Last Hardened Time

Last Redone LSN

Last Redone Time

Reset view

th
ts

den
mns

System Views

- System catalog views
- View system configuration
- Dynamic Management Views (DMVs) and Functions
- View current state and health information

```
SELECT ar.replica_server_name AS 'Replica Instance', dr_state.database_id as 'Database ID',  
ar_state.connected_state_desc AS 'Connection State', dr_state.synchronization_state_desc AS 'Synchronization State',  
dr_state.synchronization_health_desc AS 'Health State'  
FROM (( sys.availability_groups AS ag JOIN sys.availability_replicas AS ar ON ag.group_id = ar.group_id )  
JOIN sys.dm_hadr_availability_replica_states AS ar_state ON ar.replica_id = ar_state.replica_id)  
JOIN sys.dm_hadr_database_replica_states dr_state on  
ag.group_id = dr_state.group_id and dr_state.replica_id = ar_state.replica_id;
```

	Replica Instance	Database ID	Connection State	Synchronization State	Health State
1	ALWAYSONN1	5	CONNECTED	SYNCHRONIZED	HEALTHY
2	ALWAYSONN1	6	CONNECTED	SYNCHRONIZED	HEALTHY
3	ALWAYSONN2	5	CONNECTED	SYNCHRONIZED	HEALTHY
4	ALWAYSONN2	6	CONNECTED	SYNCHRONIZED	HEALTHY
5	ALWAYSONN3	5	CONNECTED	NOT SYNCHRONIZING	NOT_HEALTHY
6	ALWAYSONN3	6	CONNECTED	SYNCHRONIZING	HEALTHY

Performance Monitor

SQL Server: Availability Replica

- Bytes Received from Replica/sec
- Bytes Sent to Replica/sec
- Bytes Sent to Transport/sec
- Resent Messages/sec

SQL Server: Database Replica

- Log Send Queue
- Transaction Delay
- Mirrored Write Transactions/sec
- Redone Bytes/sec
- Redo Bytes Remaining
- Redo Blocked/sec

SQL Server: Databases

- Log Bytes Flushed/sec

**Watch for HADR_
waits, Network
Latency, WRITELOG,
and PAGEIOLATCH_
waits**

Automated Health Monitoring using PowerShell

- Cmdlets to monitor health of an Availability Group
 - Test-SQLAvailabilityGroup
 - Test-SQLAvailabilityReplica
 - Test-SQLDatabaseReplicaState

```
PS SQLSERVER:\SQL\AlwaysOnN1\Default\AvailabilityGroups\AGCorp\availabilityreplicas> DIR | Test-SQLAvailabilityReplica
```

HealthState	AvailabilityGroup	Name
Healthy	AGCorp	ALWAYSONN1
Healthy	AGCorp	ALWAYSONN2
Healthy	AGCorp	ALWAYSONN3
Healthy	AGCorp	ALWAYSONN2
Healthy	AGCorp	ALWAYSONN3
Healthy	AGCorp	Adventureworks2014
Healthy	AGCorp	AdventureworksDW2014

Demo

AlwaysOn Availability Groups

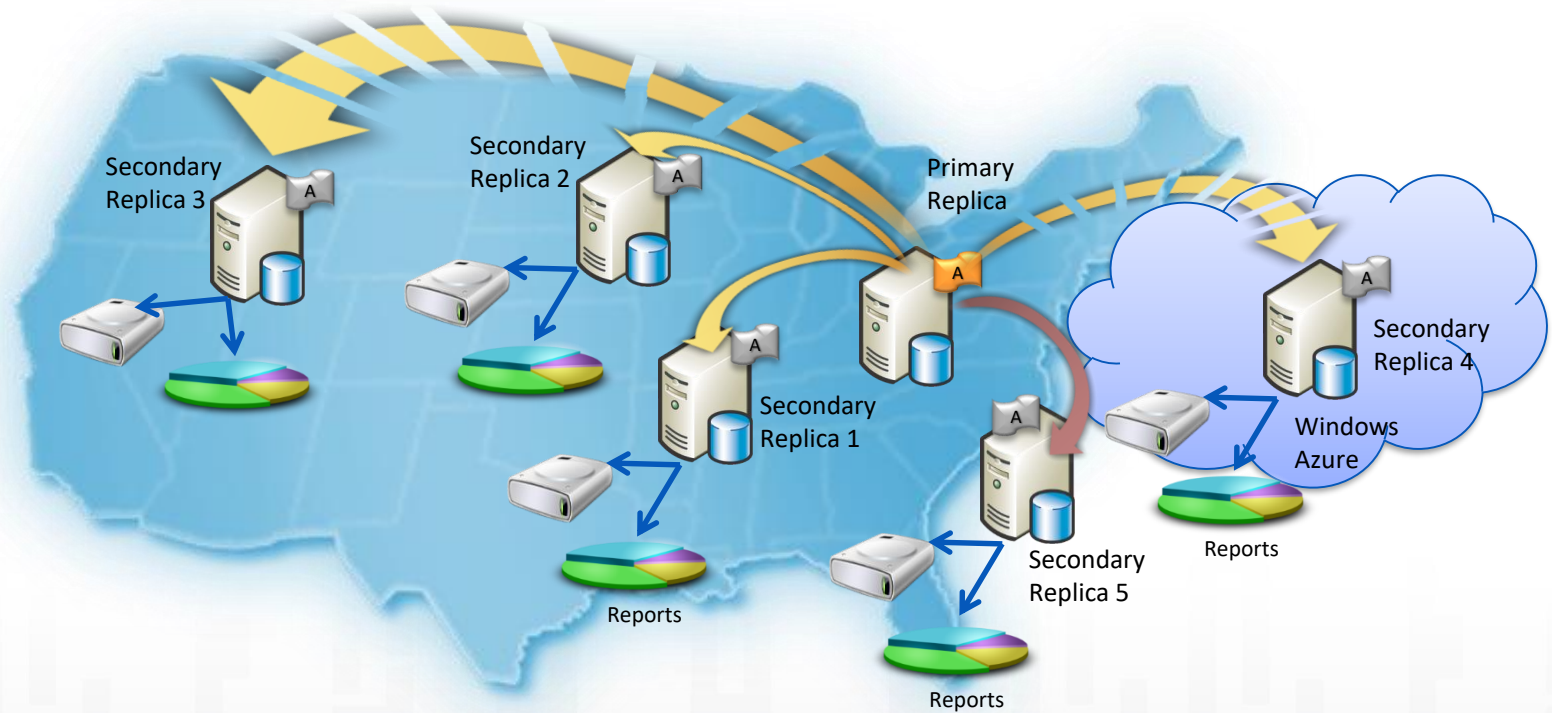
- AlwaysOn Availability Groups Health with the vw_AlwaysOn_Health view

AlwaysOn Availability Groups in SQL Server 2016

AlwaysOn Availability Groups
and Replicas



Availability Groups in SQL Server



Asynchronous Movement

Synchronous Movement



Primary Replica



Secondary Replica



Backups



Reports

AlwaysOn AG Enhancements in SQL 2016

Load balancing for secondary replicas

- Define a load balanced routing list (SSMS)

Increased Replicas for automatic failover

- Now you can have 3 replicas
- Requires synchronous commit

Group Managed Service Accounts are allowed for AG Clusters

- Managed service accounts that extend to multiple machines

Distributed Transactions are Supported

- Requires Windows Server 2016
- Use the new WITH DTC_SUPPORT for the CREATE AG command (SSMS)

Improved log performance

- Availability Groups have affectedly better performance over the wire

Database Level Failover

- If a DB goes offline you will be able to initiate failover for the AG

Basic Availability Group

- An option for Standard Edition customers in SQL Server 2016

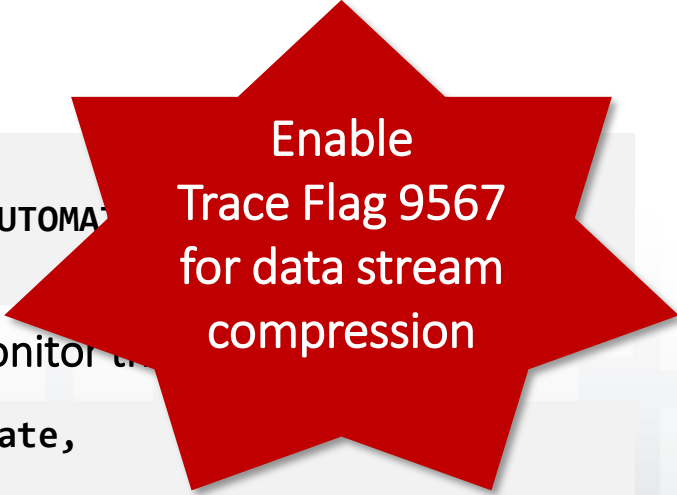
AlwaysOn AG Automatic Seeding

- Automatically create the database replica on the secondary server
- SQL Server performs a backup over the network for initialization
- Process can be slow if databases are very large, or the replica(s) are remote
- The transaction log for these databases cannot be truncated during the backup process
- Create new or enable on an Existing AG

```
ALTER AVAILABILITY GROUP [<availability_group_name>]  
MODIFY REPLICA ON '<primary_node>' WITH (SEEDING_MODE = AUTOMATIC)  
GO
```

- Query sys.dm_hadr_automatic_seeding system view to monitor the

```
SELECT start_time, completion_time, is_source, current_state,  
failure_state, failure_state_desc, error_code  
FROM sys.dm_hadr_automatic_seeding
```



Enable
Trace Flag 9567
for data stream
compression

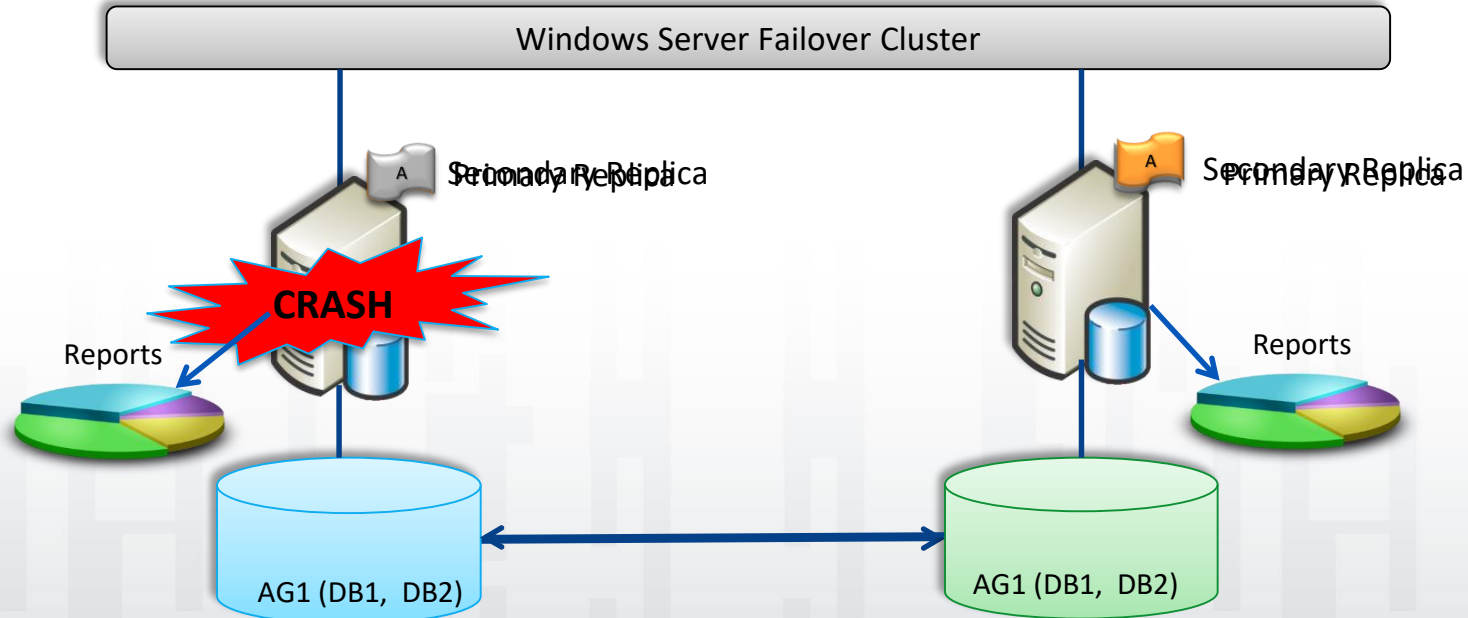
AlwaysOn Availability Group Read-Only Routing

AlwaysOn Availability Groups
and Replicas



What is Read-Only Routing?

- Facilitates automatic redirect of read workloads
- Increases availability for read workloads
- Minimizes contention on Primary Replica
- Requires Listener and ApplicationIntent property



Load Balancing in Readable Secondaries

- In SQL 2014, read-only transactions routed by the Listener, went to the first secondary that was available.
- Read-only Routing lists
- Now you can configure the ROR lists to round-robin among a specific set of secondaries. (for each primary)
- `READ_ONLY_ROUTING_LIST =
(('COMPUTER2', 'COMPUTER3', 'COMPUTER4'), 'COMPUTER5')`

Readable Secondary Load Balancing

DR Site

```
READ_ONLY_ROUTING_LIST=  
(( 'COMPUTER2', 'COMPUTER3',  
  'COMPUTER4'), 'COMPUTER5')
```

Computer 5



Computer 2



Computer 3



Computer 4



Primary Site

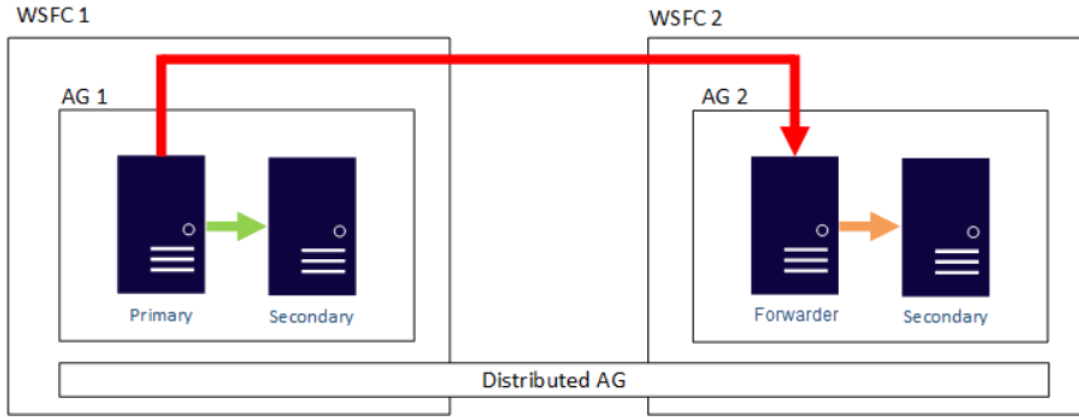
Computer 1
(Primary)



Demo

Readable Routing & Reporting with SQL Server 2016 Availability Groups

Distributed Availability Groups



Key Scenarios:

- Disaster recovery and multi-site scenarios
- Migrate by using a distributed availability group
- Scale out readable replicas with distributed availability groups

- Span geographically dispersed sites
- DR Site does not impact primary
- Secondary AG automatically becomes ReadOnly
- Replicate between AGs once
- The Primary of AG1 sends to the secondary of AG1 and the Primary of AG2 (forwarder)
- Manual failover only

SQL Server 2017 What's New with AlwaysOn and Linux

Performance and Availability



What's New in SQL Server 2017

- New **CLUSTER_TYPE** that identifies the type of server cluster manager that manages an availability group. (WSFC, EXTERNAL, NONE)
- Guarantee commits on synchronous secondary replicas
- **Read-scale availability groups** for performance
- **Cross database transactions** are now supported among all databases that are part of an Always On Availability Group
- **Transaction log redo** of memory-optimized tables is now executed in parallel. This bolsters faster recovery times and significantly increases the sustained throughput of AlwaysOn availability group configuration
- Improved **Monitoring** and **Diagnostics**

AlwaysOn AGs on Linux Concepts

High-level
Steps to
create an
availability
group on
Linux servers
for HA:

[Create Linux Cluster](#)

[Configure SQL Server on the Cluster Nodes](#)

[Create the Availability Group](#)

Configure a cluster resource
manager, like Pacemaker



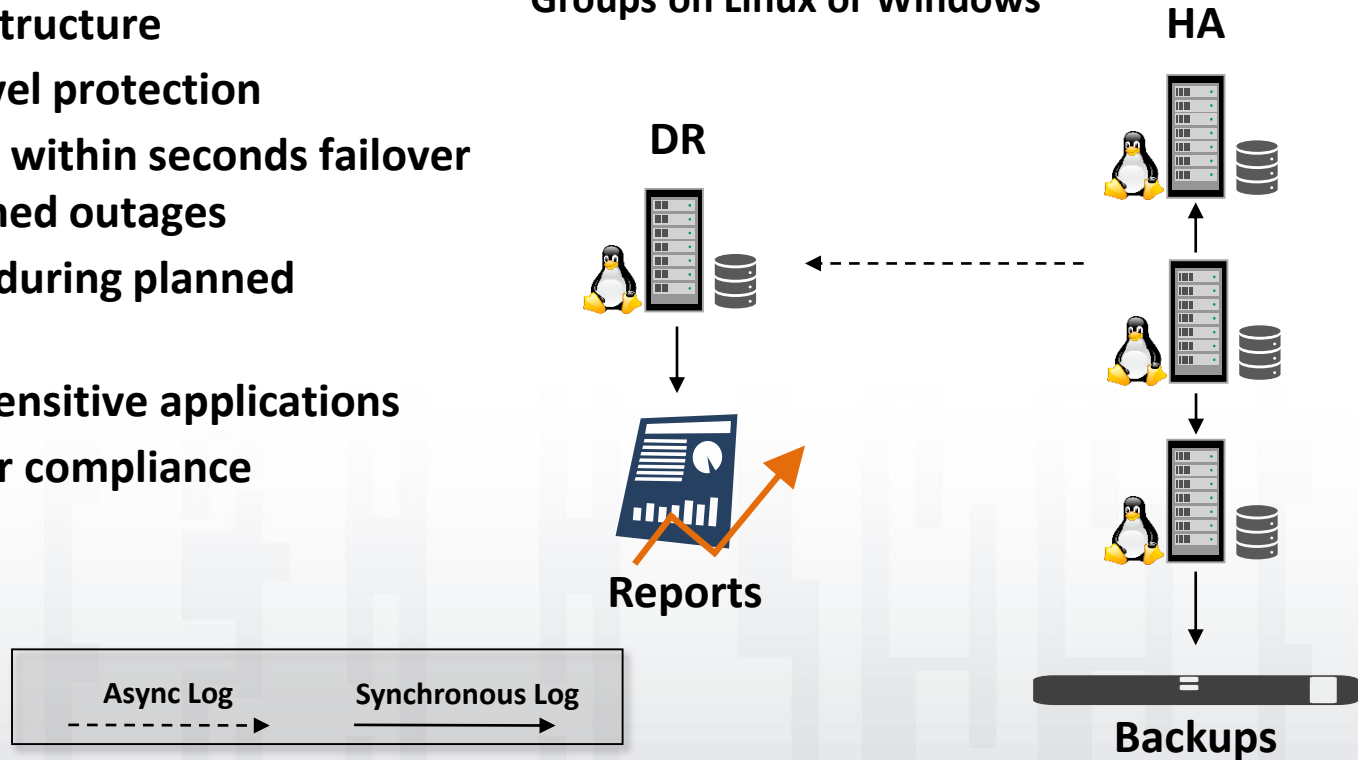
Enterprise Building a Mission Critical Application

Scenario

- All Linux Infrastructure
- Application-level protection
- Automatic and within seconds failover during unplanned outages
- No Downtime during planned Maintenance
- Performance sensitive applications
- DR required for compliance regulations

Solution

- HADR with Always On Availability Groups on Linux or Windows



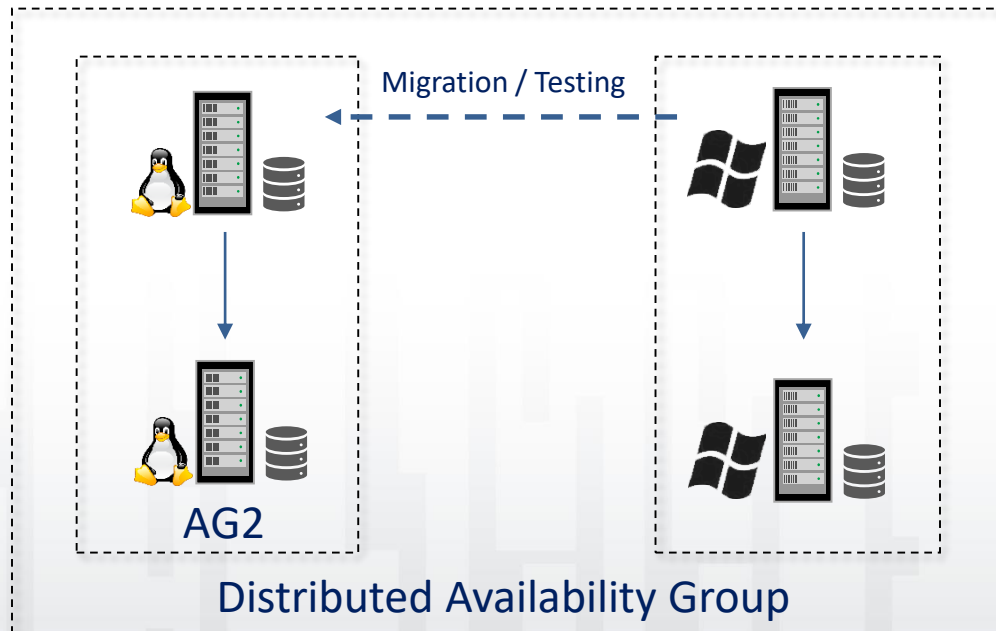
Migration / Testing

Scenario

- **ISV solution built on SQL Server on Windows**
 - Linux Certification
- **Enterprise moving to an all-Linux infrastructure**
 - Rigorous Business Requirements
 - Seamless migration
- **No cross platform cluster manager – DR Solution between environments**

Solution

- **Minimum downtime *and* HA for cross-platform migrations with Distributed Availability Groups**



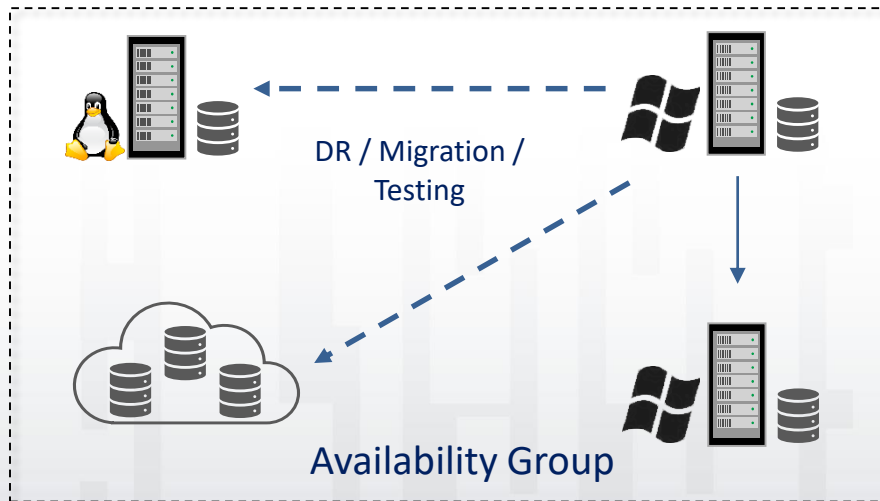
Flexibility of HADR Infrastructure

Scenario

- ISV solution built on SQL Server on Windows
 - Linux Certification

Solution

- Cross platform hybrid availability group
- Enables DR/Testing/Migration – Fast Data replication and minimum downtime
- Cost effective – Hybrid with Azure replicas



Read Scale-Out

Scenario

- SaaS app (website)
- Catalog database with high volume of concurrent read-only transactions
- Bottlenecks on primary due to read workloads
- Increased response time

Note:

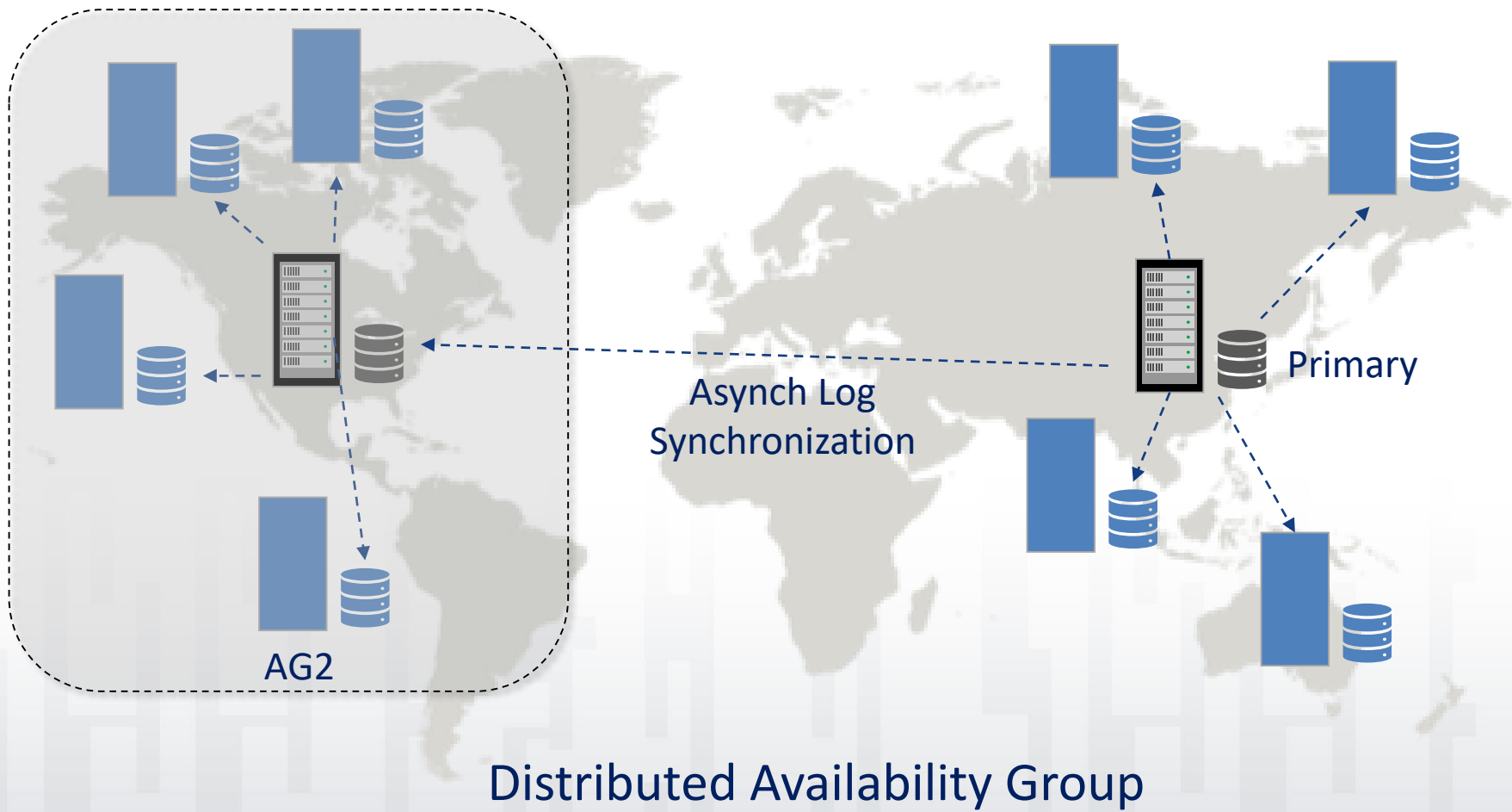
- A single distributed availability group can have up to 17 readable secondary replicas
- Daisy Chain is supported

Solution

- Read scale with availability groups
- No cluster required
- Both Windows and Linux support



Scale with Distributed Availability Groups



AlwaysOn and SQL on Linux Resources

SQL Server Linux Stability

<https://docs.microsoft.com/en-us/sql/linux/sql-server-linux-business-continuity-dr>

Install SQL Server on Linux
Quick Starts

<https://docs.microsoft.com/en-us/sql/linux/quickstart-install-connect-suse>

SQL Server Best Practices

<http://technet.microsoft.com/en-us/sqlserver/bb671430>

Always On Availability
Groups (SQL Server)

<https://docs.microsoft.com/en-us/sql/database-engine/availability-groups/windows/always-on-availability-groups-sql-server>

SQL Server Hands On Labs

<https://technet.microsoft.com/en-us/virtuallabs>

SQL Server Whitepapers

[https://msdn.microsoft.com/en-us/library/mt803150\(v=sql.1\).aspx](https://msdn.microsoft.com/en-us/library/mt803150(v=sql.1).aspx)

Perf BP and config guidelines for
SQL Server 2017 on Linux

<https://docs.microsoft.com/en-us/sql/linux/sql-server-linux-performance-best-practices>

Overview

- **Avoiding Downtime and Data Loss**
 - Two Principle Requirements for HA and DR
- **HADR Options in SQL Server**
- **AlwaysOn Availability Groups and Availability Modes**
- **Flexible Failover Policy and Health Monitoring**
- **AlwaysOn Availability Group Monitoring**
- **AlwaysOn Availability Group Features in SQL Server 2016 / 2017**
 - AlwaysOn Availability Group Read-Only Routing
 - Linux and Pacemaker Support
 - Read Scale-Out Architecture

Questions?



Don't forget to complete an online evaluation!

SQL SERVER 2016/2017 ALWAYS ON AVAILABILITY GROUPS: WINDOWS AND LINUX DEPLOYMENTS

Your evaluation helps organizers build better conferences
and helps speakers improve their sessions.

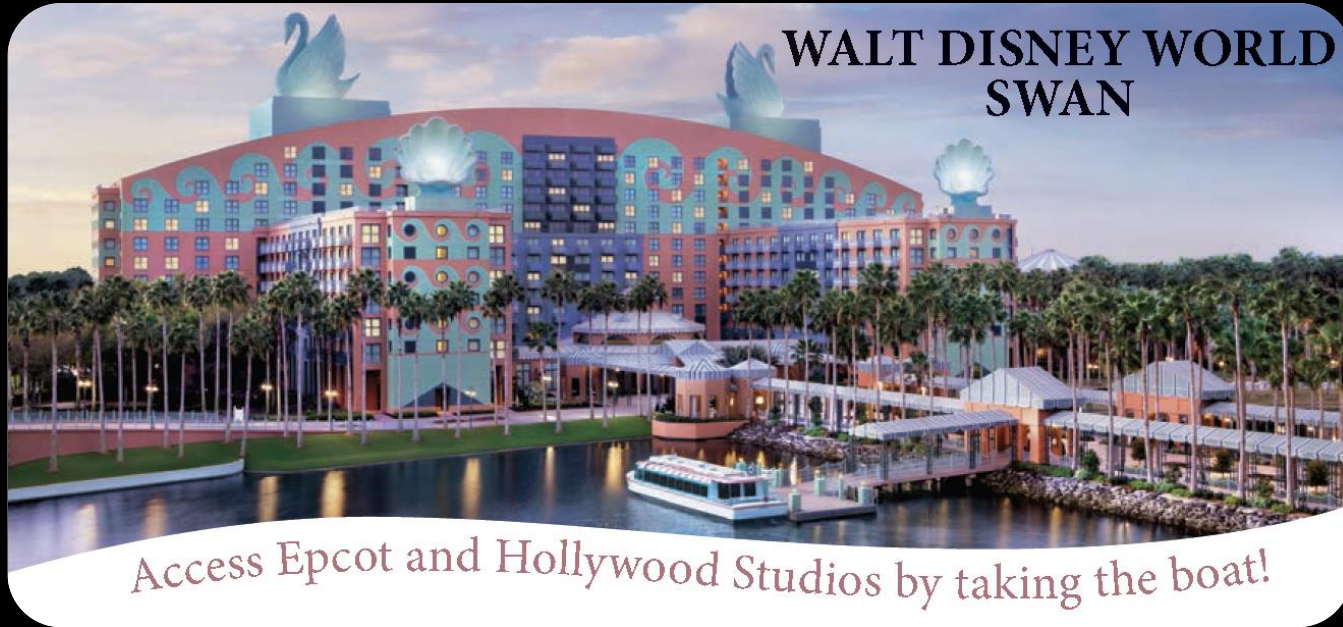


SQL
intersection

Thank you!

Save the Date!

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2018

Mar 25-28

We're back in Orlando!



Leave the every day behind and enter a world of wonder and enchantment at the Walt Disney World® Resort. Located in the heart of the most magical place on earth, the Walt Disney World Swan and Dolphin Resort provides a truly extraordinary backdrop for our event! Beautiful tropical landscaping, tranquil waterways, and classic art and architecture work together to create a stunning landmark!